

ABURATSU, Japan

WMO: 478350

Lat: 31.5786N

Lon: 131.4078E

Elev: 15

StdP: 101.15

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.4	2.5	-8.0	1.9	5.1	-6.4	2.2	5.6	12.8	9.5	11.4	10.5	4.5	290	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	32.8	25.9	31.8	26.0	31.0	25.9	27.2	30.5	26.7	30.1	26.3	29.6	5.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.8	29.0	25.8	21.2	28.7	25.4	20.6	28.5	86.0	30.6	84.0	30.3	82.2	29.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.6	9.6	8.3	-0.9	34.9	0.9	1.1	-1.6	35.7	-2.1	36.4	-2.6	37.0	-3.3	37.8
			-3.1	27.8	1.0	0.5	-3.8	28.2	-4.4	28.5	-5.0	28.8	-5.8	29.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	9.0	10.1	12.8	16.8	20.6	23.5	27.5	28.1	25.6	21.2	16.1	11.1
	DBStd	7.06	2.80	3.28	3.02	2.61	1.91	2.34	1.78	1.31	2.08	2.53	3.18	3.14
	HDD10.0	121	53	35	8	0	0	0	0	0	0	0	1	24
	HDD18.3	1066	289	231	172	59	3	0	0	0	5	81	226	
	CDD10.0	3258	22	38	96	204	329	404	542	562	469	349	183	58
	CDD18.3	1159	0	1	1	13	74	154	284	304	219	96	14	0
	CDH23.3	9733	0	0	1	26	223	859	2979	3406	1834	384	21	0
CDH26.7	2902	0	0	0	2	18	165	988	1239	456	34	1	0	
Wind	WSAvg	4.2	4.6	4.4	4.4	4.3	3.9	3.8	3.8	3.9	4.0	4.1	4.2	4.4
Precipitation	PrecAvg	2610	84	119	188	225	233	498	288	271	321	182	118	81
	PrecMax	4220	254	302	368	452	466	867	1048	654	746	493	363	216
	PrecMin	2023	27	38	86	54	121	238	41	27	69	17	32	2
	PrecStd	474	51	74	72	98	84	166	226	156	193	115	84	57
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.3	21.5	22.7	26.0	28.7	31.8	34.3	34.6	32.6	29.0	25.4	21.2
		MCWB	15.0	17.2	16.1	17.9	19.7	25.2	25.9	26.2	25.4	23.5	21.5	16.8
	2%	DB	17.3	19.2	20.9	23.7	26.8	30.1	32.8	33.0	31.0	27.6	23.7	19.4
		MCWB	13.7	16.0	15.8	18.0	20.1	24.8	25.9	26.1	25.5	22.9	19.2	15.5
	5%	DB	15.8	17.7	19.6	22.5	25.5	28.4	31.8	32.0	30.0	26.4	22.6	18.0
		MCWB	11.9	14.4	15.1	17.6	19.8	24.2	26.0	26.3	25.0	21.9	18.3	14.1
	10%	DB	14.3	16.0	18.4	21.5	24.5	27.2	30.8	31.1	29.1	25.5	21.3	16.6
		MCWB	10.4	12.3	14.4	16.9	19.6	23.8	25.8	26.1	24.5	21.0	17.4	12.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	18.1	18.9	21.5	23.7	26.1	27.7	27.8	27.2	25.5	22.2	18.6
		MCDB	18.5	20.3	20.7	23.6	26.0	29.4	31.2	31.1	30.0	27.8	24.5	20.0
	2%	WB	14.8	16.7	17.5	20.3	22.7	25.5	27.1	27.3	26.5	24.3	20.6	16.5
		MCDB	16.7	19.0	19.6	22.1	24.6	28.6	30.5	30.7	29.3	26.6	22.9	18.5
	5%	WB	12.8	14.9	16.3	19.1	21.8	25.0	26.6	26.9	25.9	23.3	19.4	14.8
		MCDB	15.0	17.3	18.8	21.4	23.7	27.7	30.0	30.3	28.7	25.5	21.6	17.3
	10%	WB	10.8	13.0	15.1	18.1	20.9	24.4	26.1	26.5	25.4	22.1	18.2	13.3
		MCDB	13.6	15.3	17.7	20.6	23.2	26.9	29.5	29.9	28.2	24.5	20.5	15.9
Mean Daily Temperature Range	5% DB	MDBR	8.3	8.2	8.1	7.9	7.2	5.4	5.7	5.9	6.0	6.8	7.7	8.4
		MCDBR	9.7	9.0	9.4	9.3	8.8	6.8	7.2	7.2	6.7	7.2	7.8	9.0
		MCWBR	7.0	6.9	6.4	5.8	5.1	3.2	2.9	3.1	3.3	4.1	5.1	6.2
	5% WB	MCDBR	8.1	8.1	7.7	6.7	6.0	5.6	6.2	6.1	5.7	5.5	6.1	7.6
		MCWBR	7.1	6.9	6.4	5.2	4.5	2.9	3.0	3.0	3.1	3.7	4.8	6.2
Clear-Sky Solar Irradiance	taub	0.426	0.472	0.534	0.555	0.564	0.565	0.543	0.531	0.517	0.491	0.441	0.413	
	taud	2.109	2.012	1.863	1.862	1.909	1.984	2.077	2.102	2.104	2.096	2.158	2.162	
	Ebn at Noon	771	772	753	757	753	749	764	768	761	746	751	761	
	Edh at Noon	130	158	196	204	196	181	164	158	152	143	123	118	
All-Sky Solar Radiation	RadAvg	2.71	3.23	4.14	4.73	4.83	3.89	5.33	5.31	4.30	3.53	2.77	2.52	
	RadStd	0.20	0.29	0.29	0.49	0.54	0.65	0.60	0.35	0.48	0.41	0.28	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page